

THE STRUCTURE OF MATHEMATICAL REALITY

Phase III — Chapter 10

Dynamics and Approximation: The Structure of Change

Exercises

Self-Study Curriculum

1. Take the logistic equation $dN/dt = rN(1 - N/K)$. Find its fixed points. Linearize around each one and use the sign of the derivative to classify its stability. Which fixed point is the carrying capacity, and why is it stable? Sketch the flow on the N -axis (arrows left/right).

2. A system you model has a “threshold”: below some input it does essentially nothing, above it the response switches on sharply. Explain why a linear model *structurally cannot* represent this, and name the dynamical phenomenon (from the lecture) that can. What would you have to measure to locate the threshold?

3. You linearize a nonlinear system around an equilibrium and the eigenvalues of the Jacobian come out as a complex-conjugate pair with a small positive real part. Describe, in words, what the system does near that equilibrium. What kind of bifurcation is the system likely near, and what would you expect to see if you nudged the parameter further?

4. Chaos and noise both look “random” in a time series. State one operational way you might

tell a low-dimensional chaotic signal from a genuinely stochastic one. Why does the fact that a chaotic system is *deterministic* not make it *predictable*?
